

Installation Guide

A CVAT installation guide for different operating systems.

Quick installation guide

Before you can use CVAT, you'll need to get it installed. The document below contains instructions for the most popular operating systems. If your system is not covered by the document it should be relatively straightforward to adapt the instructions below for other systems.

Probably you need to modify the instructions below in case you are behind a proxy server. Proxy is an advanced topic and it is not covered by the guide.

For access from China, read [sources for users from China](#) section.

Ubuntu 22.04/20.04 (x86_64/amd64)

- Open a terminal window. If you don't know how to open a terminal window on Ubuntu please read [the answer](#).
- Type commands below into the terminal window to install Docker and Docker Compose. More instructions can be found [here](#).

```
sudo apt-get update
sudo apt-get --no-install-recommends install -y \
  apt-transport-https \
  ca-certificates \
  curl \
  gnupg-agent \
  software-properties-common
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -
sudo add-apt-repository \
  "deb [arch=amd64] https://download.docker.com/linux/ubuntu \
  $(lsb_release -cs) \
  stable"
sudo apt-get update
sudo apt-get --no-install-recommends install -y \
  docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

[Copy](#)

- (Optional) To avoid prefacing Docker commands with `sudo`, you can perform the [post-installation steps](#). This involves creating a Unix group named `docker` and adding your current user to this group.

```
sudo groupadd docker
sudo usermod -aG docker $USER
```

[Copy](#)

Log out and log back in (or reboot) so that your group membership is re-evaluated. You can type `groups` command in a terminal window after that and check if `docker` group is in its output.

- Clone CVAT source code from the [GitHub repository](#) with Git.

Following command will clone the latest develop branch:

```
git clone https://github.com/cvat-ai/cvat
cd cvat
```

[Copy](#)

```
export CVAT_HOST=FQDN_or_YOUR-IP-ADDRESS
```

Copy

- Run docker containers. It will take some time to download the latest CVAT and other required images like postgres, redis, and start containers.

```
docker compose up -d
```

Copy

- (Optional) Use `CVAT_VERSION` environment variable to specify the version of CVAT you want to install specific version (e.g `v2.1.0` , `dev`). Default behavior: `dev` images will be pulled for develop branch, and corresponding release images for release versions.

```
CVAT_VERSION=dev docker compose up -d
```

Copy

- Alternative: if you want to build the images locally with unreleased changes see [How to pull/build/update CVAT images section](#)
- You can register a user but by default, it will not have rights even to view the list of tasks. Thus you should create a superuser. The superuser can use an admin panel to assign the correct groups to the user. Please use the command below:

```
docker exec -it cvat_server bash -ic 'python3 ~/manage.py createsuperuser'
```

Copy

Choose a username and a password for your admin account. For more information please read [Django documentation](#).

- Google Chrome is the only browser that is supported by CVAT. You need to install it as well. Type commands below in a terminal window:

```
curl https://dl-ssl.google.com/linux/linux_signing_key.pub | sudo apt-key add -
sudo sh -c 'echo "deb [arch=amd64] http://dl.google.com/linux/chrome/deb/ stable main" >> /etc/apt/sources.list.d/google.list'
sudo apt-get update
sudo apt-get --no-install-recommends install -y google-chrome-stable
```

Copy

- Open the installed Google Chrome browser and go to [localhost:8080](#). Type your login/password for the superuser on the login page and press the *Login* button. Now you should be able to create a new annotation task. Please read the [CVAT manual](#) for more details.

Windows 10

- Install WSL2 (Windows subsystem for Linux) refer to [this official guide](#). WSL2 requires Windows 10, version 2004 or higher. After installing WSL2, install a Linux Distribution of your choice.
- Download and install [Docker Desktop for Windows](#). Double-click `Docker for Windows Installer` to run the installer. More instructions can be found [here](#). Official guide for docker WSL2 backend can be found [here](#). Note: Check that you are specifically using WSL2 backend for Docker.
- In Docker Desktop, go to `Settings >> Resources >> WSL Integration` , and enable integration with the Linux Distribution that you chose.
- Download and install [Git for Windows](#). When installing the package please keep all options by default. More information about the package can be found [here](#).
- Download and install [Google Chrome](#). It is the only browser which is supported by CVAT.
- Go to windows menu, find the Linux distribution you installed and run it. You should see a terminal window.
- Clone CVAT source code from the [GitHub repository](#).

The following command will clone the latest develop branch:

See [alternatives](#) if you want to download one of the release versions.

- Run docker containers. It will take some time to download the latest CVAT release and other required images like postgres, redis, etc. from DockerHub and create containers.

```
docker compose up -d
```

[Copy](#)

- (Optional) Use `CVAT_VERSION` environment variable to specify the version of CVAT you want to install specific version (e.g `v2.1.0` , `dev`). Default behavior: `dev` images will be pulled for develop branch, and corresponding release images for release versions.

```
CVAT_VERSION=dev docker compose up -d
```

[Copy](#)

- Alternative: if you want to build the images locally with unreleased changes see [How to pull/build/update CVAT images section](#)
- You can register a user but by default, it will not have rights even to view the list of tasks. Thus you should create a superuser. A superuser can use an admin panel to assign correct groups to other users. Please use the command below:

```
sudo docker exec -it cvat_server bash -ic 'python3 ~/manage.py createsuperuser'
```

[Copy](#)

If you don't have winpty installed or the above command does not work, you may also try the following:

```
# enter docker image first
docker exec -it cvat_server /bin/bash
# then run
python3 ~/manage.py createsuperuser
```

[Copy](#)

Choose a username and a password for your admin account. For more information please read [Django documentation](#).

- Open the installed Google Chrome browser and go to [localhost:8080](#). Type your login/password for the superuser on the login page and press the *Login* button. Now you should be able to create a new annotation task. Please read the [CVAT manual](#) for more details.

Mac OS Mojave

- Download [Docker for Mac](#). Double-click Docker.dmg to open the installer, then drag Moby the whale to the Applications folder. Double-click Docker.app in the Applications folder to start Docker. More instructions can be found [here](#).
- There are several ways to install Git on a Mac. The easiest is probably to install the Xcode Command Line Tools. On Mavericks (10.9) or above you can do this simply by trying to run git from the Terminal the very first time.

```
git --version
```

[Copy](#)

If you don't have it installed already, it will prompt you to install it. More instructions can be found [here](#).

- Download and install [Google Chrome](#). It is the only browser which is supported by CVAT.
- Open a terminal window. The terminal app is in the Utilities folder in Applications. To open it, either open your Applications folder, then open Utilities and double-click on Terminal, or press Command - spacebar to launch Spotlight and type "Terminal," then double-click the search result.
- Clone CVAT source code from the [GitHub repository](#) with Git.

```
git clone https://github.com/cvat-ai/cvat
cd cvat
```

Copy

See [alternatives](#) if you want to download one of the release versions or use the `wget` or `curl` tools.

- Run docker containers. It will take some time to download the latest CVAT release and other required images like postgres, redis, etc. from DockerHub and create containers.

```
docker compose up -d
```

Copy

- (Optional) Use `CVAT_VERSION` environment variable to specify the version of CVAT you want to install specific version (e.g `v2.1.0` , `dev`). Default behavior: `dev` images will be pulled for develop branch, and corresponding release images for release versions.

```
CVAT_VERSION=dev docker compose up -d
```

Copy

- Alternative: if you want to build the images locally with unreleased changes see [How to pull/build/update CVAT images section](#)
- You can register a user but by default, it will not have rights even to view the list of tasks. Thus you should create a superuser. A superuser can use an admin panel to assign correct groups to other users. Please use the command below:

```
docker exec -it cvat_server bash -ic 'python3 ~/manage.py createsuperuser'
```

Copy

Choose a username and a password for your admin account. For more information please read [Django documentation](#).

- Open the installed Google Chrome browser and go to [localhost:8080](#). Type your login/password for the superuser on the login page and press the *Login* button. Now you should be able to create a new annotation task. Please read the [CVAT manual](#) for more details.

Advanced Topics

How to get CVAT source code

Git (Linux, Mac, Windows)

1. Install Git on your system if it's not already installed
 - Ubuntu:

```
sudo apt-get --no-install-recommends install -y git
```

Copy

- Windows: Follow instructions from <https://git-scm.com/download/win>
2. Clone CVAT source code from the [GitHub repository](#).

The command below will clone the default branch (develop):

```
git clone https://github.com/cvat-ai/cvat
cd cvat
```

Copy

To clone specific tag, e.g. v2.1.0:

Wget (Linux, Mac)

To download latest develop branch:

```
wget https://github.com/cvat-ai/cvat/archive/refs/heads/develop.zip
unzip develop.zip && mv cvat-develop cvat
cd cvat
```

Copy

To download specific tag:

```
wget https://github.com/cvat-ai/cvat/archive/refs/tags/v1.7.0.zip
unzip v1.7.0.zip && mv cvat-1.7.0 cvat
cd cvat
```

Copy

Curl (Linux, Mac)

To download the latest develop branch:

```
curl -LO https://github.com/cvat-ai/cvat/archive/refs/heads/develop.zip
unzip develop.zip && mv cvat-develop cvat
cd cvat
```

Copy

To download specific tag:

```
curl -LO https://github.com/cvat-ai/cvat/archive/refs/tags/v1.7.0.zip
unzip v1.7.0.zip && mv cvat-1.7.0 cvat
cd cvat
```

Copy

CVAT healthcheck command

The following command allows testing the CVAT container to make sure it works.

```
docker exec -t cvat_server python manage.py health_check
```

Copy

The expected output of a healthy CVAT container:

```
Cache backend: default    ... working
DatabaseBackend           ... working
DiskUsage                 ... working
MemoryUsage              ... working
MigrationsHealthCheck    ... working
OPAHealthCheck           ... working
```

Copy

Deploying CVAT behind a proxy

home directory of the user which starts containers and add json such as the following:

```
{
  "proxies": {
    "default": {
      "httpProxy": "http://proxy_server:port",
      "httpsProxy": "http://proxy_server:port",
      "noProxy": "*,test.example.com,example2.com"
    }
  }
}
```

Copy

These environment variables are set automatically within any container. Please see the [Docker documentation](#) for more details.

Using the Traefik dashboard

If you are customizing the docker compose files and you come upon some unexpected issues, using the Traefik dashboard might be very useful to see if the problem is with Traefik configuration, or with some of the services.

You can enable the Traefik dashboard by uncommenting the following lines from `docker-compose.yml`

```
services:
  traefik:
    # Uncomment to get Traefik dashboard
    # - "--entryPoints.dashboard.address=:8090"
    # - "--api.dashboard=true"
    # labels:
    # - traefik.enable=true
    # - traefik.http.routers.dashboard.entrypoints=dashboard
    # - traefik.http.routers.dashboard.service=api@internal
    # - traefik.http.routers.dashboard.rule=Host(`${CVAT_HOST:-localhost}`)
```

Copy

and if you are using `docker-compose.https.yml`, also uncomment these lines

```
services:
  traefik:
    command:
      # Uncomment to get Traefik dashboard
      # - "--entryPoints.dashboard.address=:8090"
      # - "--api.dashboard=true"
```

Copy

Note that this “insecure” dashboard is not recommended in production (and if your instance is publicly available); if you want to keep the dashboard in production you should read Traefik’s [documentation](#) on how to properly secure it.

Additional components

Semi-automatic and automatic annotation

Please follow this [guide](#).

Stop all containers

The command below stops and removes containers and networks created by `up`.

Use your own domain

If you want to access your instance of CVAT outside of your localhost (on another domain), you should specify the `CVAT_HOST` environment variable, like this:

```
export CVAT_HOST=<YOUR_DOMAIN>
```

Copy

Share path

You can use shared storage for uploading data when you create a task. To do that, you need to mount the shared storage to the CVAT docker container. Example of docker-compose.override.yml for this purpose:

```
services:
  cvat_server:
    volumes:
      - cvat_share:/home/django/share:ro
  cvat_worker_import:
    volumes:
      - cvat_share:/home/django/share:ro
  cvat_worker_export:
    volumes:
      - cvat_share:/home/django/share:ro
  cvat_worker_annotation:
    volumes:
      - cvat_share:/home/django/share:ro
  cvat_worker_chunks:
    volumes:
      - cvat_share:/home/django/share:ro

volumes:
  cvat_share:
    driver_opts:
      type: none
      device: /mnt/share
      o: bind
```

Copy

You can change the share device path to your actual share.

You can [mount](#) your cloud storage as a FUSE and use it later as a share.

Email verification

You can enable email verification for newly registered users. Specify these options in the [settings file](#) to configure Django allauth to enable email verification (ACCOUNT_EMAIL_VERIFICATION = 'mandatory'). Access is denied until the user's email address is verified.

```
ACCOUNT_AUTHENTICATION_METHOD = 'username_email'
ACCOUNT_CONFIRM_EMAIL_ON_GET = True
ACCOUNT_EMAIL_REQUIRED = True
ACCOUNT_EMAIL_VERIFICATION = 'mandatory'

# Email backend settings for Django
EMAIL_BACKEND = 'django.core.mail.backends.smtp.EmailBackend'
```

Copy

Deploy CVAT on the Scaleway public cloud

Please follow [this tutorial](#) to install and set up remote access to CVAT on a Scaleway cloud instance with data in a mounted object storage bucket.

Deploy secure CVAT instance with HTTPS

Using Traefik, you can automatically obtain a TLS certificate for your domain from Let's Encrypt, enabling you to use HTTPS protocol to access your website.

To enable this, first set the `CVAT_HOST` (the domain of your website) and `ACME_EMAIL` (contact email for Let's Encrypt) environment variables:

```
export CVAT_HOST=<YOUR_DOMAIN>
export ACME_EMAIL=<YOUR_EMAIL>
```

[Copy](#)

Then, use the `docker-compose.https.yml` file to override the base `docker-compose.yml` file:

```
docker compose -f docker-compose.yml -f docker-compose.https.yml up -d
```

[Copy](#)

In the firewall, ports 80 and 443 must be open for inbound connections from any

Then, the CVAT instance will be available at your domain on ports 443 (HTTPS) and 80 (HTTP, redirects to 443).

Deploy CVAT with an external database

By default, `docker compose up` will start a PostgreSQL database server, which will be used to store CVAT's data. If you'd like to use your own PostgreSQL instance instead, you can do so as follows. Note that CVAT only supports the same major version of PostgreSQL as is used in `docker-compose.yml`.

First, define environment variables with database connection settings:

```
export CVAT_POSTGRES_HOST=<PostgreSQL hostname> # mandatory
export CVAT_POSTGRES_PORT=<PostgreSQL port> # defaults to 5432
export CVAT_POSTGRES_DBNAME=<PostgreSQL database name> # defaults to "cvat"
export CVAT_POSTGRES_USER=<PostgreSQL role name> # defaults to "root"
export CVAT_POSTGRES_PASSWORD=<PostgreSQL role password> # mandatory
```

[Copy](#)

Then, add the `docker-compose.external_db.yml` file to your `docker compose up` command:

```
docker compose -f docker-compose.yml -f docker-compose.external_db.yml up -d
```

[Copy](#)

How to pull/build/update CVAT images

- **For a CVAT version lower or equal to 2.1.0**, you need to pull images using docker because the compose configuration always points to the latest image tag, e.g.

```
docker pull cvat/server:v1.7.0
docker tag cvat/server:v1.7.0 openvino/cvat_server:latest

docker pull cvat/ui:v1.7.0
docker tag cvat/ui:v1.7.0 openvino/cvat_ui:latest
```

[Copy](#)

```
CVAT_VERSION=dev docker compose pull
```

Copy

- To build images yourself include `docker-compose.dev.yml` compose config file to `docker compose` command. This can be useful if you want to build a CVAT with some source code changes.

```
docker compose -f docker-compose.yml -f docker-compose.dev.yml build
```

Copy

- To update local images to `latest` or `dev` tags run:

```
CVAT_VERSION=dev docker compose pull
```

Copy

or

```
CVAT_VERSION=latest docker compose pull
```

Copy

Troubleshooting

Sources for users from China

If you stay in China, for installation you need to override the following sources.

- For use `apt update` using:

[Ubuntu mirroring help](#)

Pre-compiled packages:

```
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal main restricted universe multiverse
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal-updates main restricted universe multiverse
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal-backports main restricted universe multiverse
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal-security main restricted universe multiverse
```

Copy

Or source packages:

```
deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal main restricted universe multiverse
deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal-updates main restricted universe multiverse
deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal-backports main restricted universe multiverse
deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ focal-security main restricted universe multiverse
```

Copy

- [Docker mirror station](#)

Add registry mirrors into `daemon.json` file:

```
{
  "registry-mirrors": [
    "http://f1361db2.m.daocloud.io",
    "https://docker.mirrors.ustc.edu.cn",
```

Copy

```
}
}
```

- For using `pip` :

[PyPI mirroring help](#)

```
pip config set global.index-url https://pypi.tuna.tsinghua.edu.cn/simple
```

- For using `npm` :

[npm mirroring help](#)

```
npm config set registry https://registry.npm.taobao.org/
```

- Instead of `git` using `gitee` :

[CVAT repository on gitee.com](#)

- For replace acceleration source `docker.com` run:

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -
sudo add-apt-repository \
    "deb [arch=amd64] https://download.docker.com/linux/ubuntu \
    $(lsb_release -cs) \
```

- For replace acceleration source `google.com` run:

```
curl https://dl-ssl.google.com/linux/linux_signing_key.pub | sudo apt-key add -
```

HTTPS is not working because of a certificate

If you're having trouble with an SSL connection, to find the cause, you'll need to get the logs from traefik by running:

```
docker logs traefik
```

The logs will help you find out the problem.

If the error is related to a firewall, then:

- Open ports 80 and 443 for inbound connections from any.
- Delete `acme.json` . The location should be something like: `/var/lib/docker/volumes/cvat_cvat_letsencrypt/_data/acme.json` .

After `acme.json` is removed, stop all cvat docker containers:

```
docker compose -f docker-compose.yml -f docker-compose.https.yml down
```

Make sure variables set (with your values):

and restart docker:

```
docker compose -f docker-compose.yml -f docker-compose.https.yml up -d
```

Copy